THE RENT-GROWTH SCREEN

Screening US Rental Markets for Future Rent Growth with Free Public Data

A Methodology and Validation Paper · Results and Negative Results Disclosed

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Abstract

Commercial real-estate research runs on expensive proprietary data. This project asks whether a disciplined screen built entirely on free public sources (Census, IRS, BLS, BEA, Zillow, FRED) can still identify, in advance, which of the 110 largest US rental markets will deliver the strongest rent growth over the following three years. The screen scores each market on eight validated measures of demand, supply, affordability, momentum, and resilience, with hand-set published weights and no statistical fitting. Walk-forward validation over 2016-2022 windows yields a pooled top-weighted rank agreement (weighted Kendall's tau) of 0.43 with realized three-year rent growth (95% CI [0.33, 0.51]); the screen's top ten out-grew the median market by +6.0 percentage points per window, and an industry-style equal-weight conditions index rebuilt from the same free data scores 0.14 on the identical task. The project's distinguishing discipline is governance: every configuration faces a pre-registered, single-attempt validation gate; five gates have run, three failed, and all outcomes are published; every published ranking is frozen to an immutable registry with its resolution dates pre-committed. This paper documents the data, the measures, the weighting argument, the validation design and results, the governance rules, and the honest limits of the approach.

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1. Introduction and Research Question

Institutional real-estate investors buy market selection: proprietary rent series, pipeline databases, and research teams that rank metros for acquisition. The premise of this project is a specific, testable question: how much of that edge is actually in the data access, and how much is in the discipline? Put concretely: can a screen built exclusively from free public sources rank the largest US rental markets by their future three-year rent growth well enough to be useful?

The claim under examination is deliberately narrow. The screen does not forecast rent levels, returns, or any single market's path; it ranks markets cross-sectionally by fundamentals that have historically preceded strong rent growth, and it is graded on whether that ranking agreed with what subsequently happened. Throughout, the object is a screen (a disciplined starting point for where to look closer), not investment advice, and the paper reports the conditions under which the screen fails as prominently as those under which it works.

The project is public by design. Every method is documented in a running decision log, every published ranking is frozen to an immutable registry before its outcome window closes, failed experiments are published alongside successes, and the dates on which the outstanding forecasts will be graded are pre-committed. The methodological posture borrows from pre-registration practice: specifications are logged before computation, each gets one attempt, and first results are final.

2. Data and the Vintage Discipline

2.1 Universe and sources

The universe is every US metropolitan area with at least 500,000 residents and continuous rent-index coverage: 110 metros. All inputs are free and public: Census population estimates, housing-unit estimates, and building permits; IRS county-to-county migration; BLS employment (QCEW and CES) and industry mix; BEA personal income; Zillow rent and home-value indices; and FRED mortgage rates. No paid source enters the model at any point.

2.2 The vintage rule

No accuracy number is reported without its data vintage. Federal metro data arrives with lags of months (permits, employment) to two years (income, migration), and revisions are common, so the project maintains two published editions at all times: a fully finalized vintage edition and a current edition built with validated fast-publishing substitutes (Section 8). Backtests additionally report a real-time variant using only inputs a user could have held at the time, alongside the finalized-data ceiling no live user ever had.

2.3 Boundary corruption and the automated quality regime

The project's hardest data lesson is on the record: the 2023 federal metro-boundary redraw (and its predecessors) silently corrupted every CBSA-keyed federal series in the panel. Agencies mixed old and new boundaries across years, manufacturing fake growth prints, and twice an artifact briefly held the #1 rank. Systematic sweeps found 35 metros requiring employment rebuilt from county files on current boundaries and 36 requiring population and housing rebuilt from county estimates; the repairs moved individual ranks substantially while headline accuracy barely moved, which is the signature of data hygiene rather than model change. Since

July 2026 a blocking quality regime cross-checks every input against an independent sister series (QCEW vs CES, ZHVI vs FHFA, ACS vs PEP), flags distributional outliers and boundary changes, asserts golden-metro regression values on every rebuild, and mechanically prevents publication until every flag is dispositioned in the public decision log.

3. The Eight Measures

Each market-year is scored on eight measures, chosen to capture the fundamentals that precede rent growth and de-duplicated after a correlation audit (an earlier ten-measure version folded population growth into migration and the multifamily pipeline into total permitting). Within each year, every measure is standardized across markets (a z-score), so nationwide swings cancel and only relative standing counts; measures where more is worse are sign-flipped; a market missing a measure receives the neutral cross-market average, disclosed wherever it occurs.

Measure	Definition	Direction
Net domestic migration	net migrants / population	higher better
Job growth	YoY total employment growth	higher better
Income growth	YoY per-capita income growth	higher better
Permits to stock	permitted units / housing stock	lower better
Rent to income	annual rent / per-capita income	lower better
Cost to own vs rent	mortgage payment on the typical home / rent	higher better
Trailing rent growth	YoY rent-index growth	higher better
Employment diversity	1 - industry concentration (HHI)	higher better

Table 1. The eight measures. YoY growth always uses the exact prior calendar year, so panel gaps never manufacture multi-year jumps.

4. Weighting: The Case Against Fitting

The composite score is a fixed weighted sum: Demand 40% (migration 20, jobs 12, income 8), Supply 25 (permits to stock, counted inversely), Affordability 20 (rent to income 12, cost to own vs rent 8), Momentum 10 (trailing rent growth), Resilience 5 (employment diversity). The weights are set by judgment, published in full, and never statistically fitted.

The no-fitting rule is a deliberate methodological position, not an omission. With roughly 110 markets and six overlapping evaluation windows, fitted weights would be overfitting by construction; the decision-science literature on improper linear models (Dawes 1979) shows that sensible fixed weights routinely match or beat fitted ones out of sample in small noisy panels. The project measured this locally: a weight-scheme study found the accuracy landscape flat across reasonable hypothesis-driven schemes, and equal weighting over the same eight measures scores within a few hundredths of the hand-set scheme. The conclusion the project draws, and states publicly, is that the weights are not the intellectual property; the component selection and the validation discipline are.

5. Validation Design

5.1 Walk-forward protocol

For each scoring year T, the screen is computed from data available for year T and compared against realized rent growth from T to T+3 (the primary horizon; one-year results are reported as a contrast). The model never sees the future; prediction years roll forward across 2016-2022, giving six completed three-year windows through 2025. Realized growth is winsorized at the 1st and 99th percentile within each window. Windows are tagged by regime (pre-COVID, the 2020-22 shock, normalization) and reported per-regime as well as pooled, because the pooled average conceals the single most decision-relevant fact about the screen (Section 6.3).

5.2 Metrics

The primary metric is top-weighted Kendall's tau, weighted by realized rank so that agreement on the true top markets counts most; 0 means no relationship, 1 perfect agreement. The headline supplementary metric is precision@10: the share of the screen's top ten that landed in the top quartile of realized growth. Because one top-10 miss moves precision@10 by ten points, precision@20 is co-reported once (0.58 pooled beside 0.65); the story is unchanged at half the lump size. For readers who think in units of money rather than rank statistics, every result is also translated into percentage points of rent growth: the average excess three-year growth of the screen's top ten over the median market.

5.3 Statistical machinery

Confidence intervals use a metro-cluster bootstrap (metros resampled with replacement, B=800-1,000, fixed seed), computed paired across competing rankings so that gaps between strategies get their own intervals. Temporal fragility is probed separately: a per-window range, a leave-one-window-out jackknife, and a state-cluster bootstrap that treats whole states as the unit of chance. All thresholds, seeds, and consequence rules are frozen in dated decision-log entries before any computation runs.

6. Validation Results

6.1 Headline accuracy

On finalized data, the pooled three-year weighted tau is **0.431** (95% metro-cluster CI [0.331, 0.509]); precision@10 is 0.65. Pre-COVID windows, the cleanest test, score 0.585 with precision@10 of 0.88. At the one-year horizon tau rises to 0.490, but the screen's purpose is the three-year horizon, where fundamentals rather than momentum carry the signal. In plain units, the screen's top ten out-grew the median market by **+6.0 percentage points** of three-year rent growth per completed window (rent momentum alone: +4.8). The real-time variant, using only data a user could have held, retains 0.42 pooled against the finalized 0.43.

6.2 Against baselines and industry practice

Ranking rule	3-yr tau	Precision@10
Full model (composite)	0.43	65%
Best single indicator	0.39	65%
Momentum only (trailing rent)	0.39	65%
Equal weight (8 ind.)	0.34	58%
Persistence (trailing h-yr)	0.22	38%
Industry-style index (equal weight)	0.14	33%
Random (avg of seeds)	0.00	26%

Table 2. All rows computed on finalized data, identical years and markets.

The most consequential row is the industry-style index: a faithful free-data replica of a leading published opportunity matrix (ten equal-weighted condition categories; six replicable from free sources), frozen before its single run. It scores 0.14 at this task, below simple persistence, and the composite's edge over it is reliable (gap +0.29, 95% CI [+0.17, +0.45]). The project's own pre-registered prediction about that replica (that it would be re-packaged rent momentum) was wrong and is published as such: its correlation with trailing rent growth is only ~0.2. The honest diagnosis is that untested conditions components dilute predictive signal. The claim is about equal-weight conditions indices at this prediction task, not about any vendor's product at its own task, and the replica omits four proprietary categories that may be its originator's strongest.

6.3 The regime caveat, stated as a result

In the 2021-22 shock windows the screen's agreement falls to 0.12 and the top-ten edge to roughly zero, while pure momentum flips firmly negative. Across the 6 observed windows tau ranges from -0.04 to +0.63; no single window drives the pooled result (jackknife range [0.39, 0.53]), and under the stricter state-cluster bootstrap the pooled interval widens to [0.34, 0.55]. A published ex-ante rule flags elevated-uncertainty scoring years (national rent growth above a fixed threshold, the only two historical firings being 2021-22) wherever rankings are shown.

7. Governance: Gates, Freezes, and Negative Results

The governance layer is the project's central methodological claim: results are only as credible as the process that could have falsified them. Five rules bind every change.

- **Pre-registration with one attempt.** No predictive computation may run on a candidate before a dated log entry freezes its definition, gate, thresholds, consequence, and attempt count; a specification logged on day T may not run before day T+1 (the cooling-off rule); at most one new attempt per federal data-release cycle; first results are final, with no re-runs and no tuning.
- **Frozen registry.** Every published run is frozen with its scores, inputs, and settings, and never edited; prior runs are annotated, never deleted; comparisons against earlier editions read from the registry so history cannot be silently rewritten.
- **Negative results publish.** Failed gates, rejected proxies, and wrong pre-registered predictions are published with the same prominence as successes.
- **Blocking data QA.** Publication is mechanically impossible until the automated quality report for the exact panel build is clean or fully dispositioned (Section 2.3).
- **Claims match evidence.** A headline claim that fails its own published sensitivity analysis may not ship; the honest object (a tier, a range) ships instead.

7.1 The gate record

#	Configuration	Retention	Top-10 overlap	Verdict
1	2025 screen, five estimated inputs	74.8%	-	Fail
2	2025 screen, fresher jobs data	84.66%	-	Fail (by 0.34)
3	2024-vintage screen, one estimated input	95.5%	8.3 / 10	Pass
4	2025 screen, state-chained income	96.6%	7.4 / 10	Pass
5	Mid-year 2026 screen, five months of data	82.7%	4.8 / 10	Fail (both bars)

Table 3. The five pre-registered configuration gates (bar: retain at least 85% of the finalized model's signal and match its top ten on at least 7 of 10 names). Attempt 2 was pulled rather than rounded up; attempt 5 ships only as a labeled speculative outlook (Section 8.2).

Separately, nine candidate measures and model variants have faced one-shot gates (five external signals including operating income and absorption indices; vacancy and unemployment changes; a three-year smoothing of the noisy growth inputs that proved reliably calmer but reliably less accurate, retention 84% with a loss CI entirely below zero). Zero were adopted; the model has remained frozen at v2.0.0 throughout, and every rejection is published with its numbers.

8. The Current Screen and the Speculative Outlook

8.1 The validated 2025-2028 screen

The slowest inputs publish one to two years late, so a current screen requires substitutes. Each substitute was validated individually against its finalized counterpart before the configuration as a whole faced a gate: Census population-estimate migration for IRS migration (rank agreement ~0.9), monthly CES employment for QCEW (0.90-0.96 per year), and the scoring-year income level chained from the prior finalized metro level by the primary state's income growth (0.60 mean rank agreement, versus 0.11 for the flat carry it replaced). This configuration is gate attempt 4 in Table 3; its pseudo-test retained 96.6% of the finalized model's signal, and the resulting 2025-2028 screen (currently led by Albany-Schenectady-Troy) is the published current edition. One disclosed asymmetry: the fix that produced attempt 4 was diagnosed on the same historical windows it was then scored on, so its genuine test is the frozen forward call (Section 11).

8.2 The speculative mid-year outlook

Reader demand for the newest possible view led to a mid-year 2026 recipe built from five months of data. Its gate (attempt 5) failed both bars, and per the consequence frozen in advance it ships only as a clearly separated speculative surface carrying its measured accuracy as a warning beside every number. A subsequent author-directed revision added a quarterly state income chain (instrument agreement 0.464, measured before adoption) and was re-measured descriptively, not re-gated: retention 89.0%, top-10 overlap 5.0/10. The surface remains speculative and cannot earn the validated label without a properly gated future specification; the episode is reported because it illustrates the governance design under product pressure: the recipe could improve, the label could not.

9. Communicating Uncertainty

Exact ranks oversell precision. The two fastest-moving inputs (job and income growth) agree between editions at Spearman 0.29 and 0.28 respectively, while every other input agrees above 0.83; measured against that noise, each market's rank is re-computed 1,000 times with those inputs jittered at their measured edition-to-edition agreement, and the published object is the 90% rank interval plus a deterministic tier (a market joins the leading cluster when its interval reaches the top ten and its median rank sits in the top quarter; 23 markets currently qualify). Editions themselves turn over: even fully finalized consecutive years historically keep only one to six of the same top-ten names, a fact stated on every ranking surface. The interval captures input-measurement noise only; model error is what the walk-forward record measures.

10. Limitations

- The rent indices measure asking rents on advertised units, not signed leases or renewals.
- No capital-markets or operating-cost data enters the model (sale prices, cap rates, insurance, taxes); rent growth stands in for profitability, and insurance-cost shocks of the kind Florida experienced move multifamily economics in ways no rent-side measure captures. Repeated searches for a free, full-coverage expense signal have failed their acquisition audits and are documented as such.

- Weights are judgment-set and stress-tested, not estimated; the defense is robustness, not optimality.
- In shock regimes the screen loses most of its edge (Section 6.3); it is a screen for normal conditions and says so wherever it is shown.
- Six overlapping windows are directional evidence, not significance; the cross-sectional intervals are conditional on the history that happened to occur.

11. The Forward Test

Backtests, however carefully walk-forwarded, share their historical sample with the researchers who designed the model. The registry is the project's only genuine out-of-sample instrument, and its scoring is pre-committed and un-skippable: the frozen 2023-vintage calls are graded when finalized 2026 rent data closes (mid-2027), and the frozen 2024-vintage and 2025-2028 screens when their windows close (2028 and early 2029 respectively), with the results published whatever they show. The resolution machinery is already built and reports, today, exactly which data each frozen run still awaits. This paper will be revised with those resolutions; per the project's rules, the revision will be disclosed, not silent.

Primary Sources

- US Census Bureau: Population Estimates Program, Housing Unit Estimates, Building Permits Survey, American Community Survey; Office of Management and Budget metro delineations.
- Internal Revenue Service: county-to-county migration flows.
- Bureau of Labor Statistics: Quarterly Census of Employment and Wages; Current Employment Statistics.
- Bureau of Economic Analysis: county and state personal income (annual and quarterly).
- Zillow Research: Observed Rent Index (ZORI) and Home Value Index (ZHVI); FHFA house-price indices (quality assurance only); Federal Reserve Economic Data: 30-year mortgage rates.
- Dawes, R. M. (1979). The robust beauty of improper linear models in decision making. American Psychologist, 34(7), 571-582.
- Arbor Realty Trust and Chandan Economics, Top Markets for Multifamily Investment Report (Spring 2026): the published industry matrix whose construction the Section 6.2 replica follows.